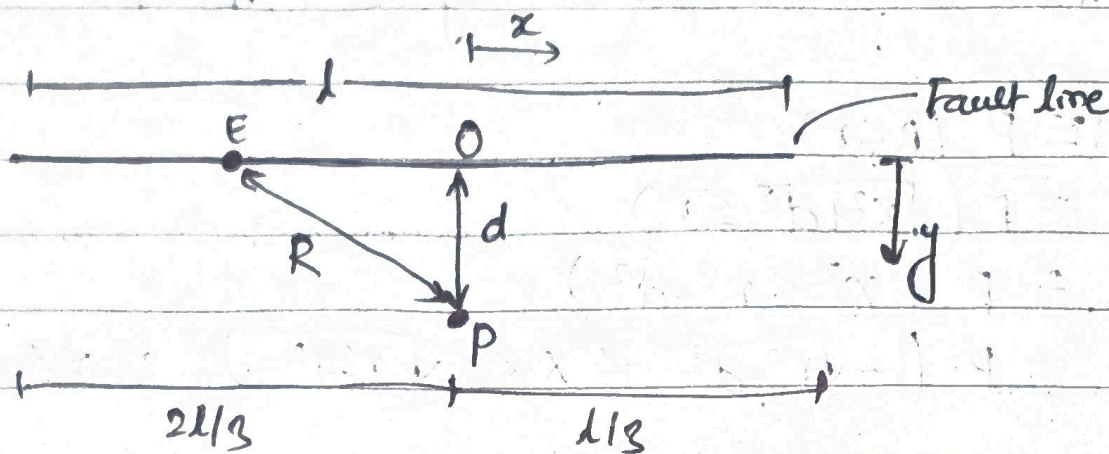


Exam: 2

1

①

→ Solⁿ.



P = power plant

E = epicenter

Given,

l = length of fault

R = distance of power plant from epicenter (Random Variable)

Let O be the origin, the horizontal axis be the x -axis and the vertical axis be the y -axis as shown.

Since the epicenter is equally likely to occur anywhere along the fault line, its probability distribution function is,

$$f(x) = \begin{cases} 1/l & -2l/3 \leq x \leq l/3 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (a)}$$

Also,

Using Pythagorean theorem,

$$R^2 = d^2 + x^2$$

$$x = \sqrt{R^2 - d^2}$$

$$R \in [d, \sqrt{(2l/3)^2 + d^2}]$$

2

(a) CDF and PDF of R,

Cumulative Distribution function of R ($F_R(r)$),

$$\begin{aligned} F_R(r) &\equiv P(R \leq r) \\ &\equiv P(\sqrt{x^2 + d^2} \leq r) \\ &\equiv P(x^2 \leq r^2 - d^2) \\ &\equiv P(-\sqrt{r^2 - d^2} \leq x \leq \sqrt{r^2 - d^2}) \quad \text{--- (b)} \end{aligned}$$

(1) At $x=0$ or $r \in [0, d)$

$$F_R(r) = 0, \text{ since } r \in [d, \infty)$$

(2) For $x \in [-1/3, 1/3]$ or $r \in [d, \sqrt{(1/3)^2 + d^2}]$:

from eqn (a) and (b),

$$F_R(r) = \int_{-\sqrt{r^2 - d^2}}^{\sqrt{r^2 - d^2}} \frac{1}{l} dx$$

$$= \frac{x}{l} \Big|_{-\sqrt{r^2 - d^2}}^{\sqrt{r^2 - d^2}}$$

$$= \frac{2\sqrt{r^2 - d^2}}{l}$$

(3) For $x \in [-2l/3, -l/3)$ or $r \in (\sqrt{(l/3)^2 + d^2}, \sqrt{(2l/3)^2 + d^2})$

from equation (a) there is no region on the right side beyond $1/3$,

$$\text{Thus, } F_R(r) = \int_{-\sqrt{r^2 - d^2}}^{1/3} \frac{1}{l} dx + \int_{1/3}^{\sqrt{r^2 - d^2}} 0 dx$$

$$\Rightarrow F_R(r) = \frac{x}{l} \Big|_{-\sqrt{r^2-d^2}}^{l/3} + 0$$

$$= \frac{1}{l} \left(\frac{l}{3} + \sqrt{r^2-d^2} \right)$$

$$= \frac{1}{3} + \frac{\sqrt{r^2-d^2}}{l}$$

Combining (1), (2) and (3),

$$F_R(r) = \begin{cases} 0 & r < d \\ \frac{2\sqrt{r^2-d^2}}{l} & d \leq r \leq \sqrt{(l/3)^2+d^2} \\ \frac{1}{3} + \frac{\sqrt{r^2-d^2}}{l} & \sqrt{(l/3)^2+d^2} < r \leq \sqrt{(2l/3)^2+d^2} \\ 1 & r > \sqrt{(2l/3)^2+d^2} \end{cases}$$

Probability Distribution Function of R ($f_R(r)$),

$$f_R(r) = \frac{d F_R(r)}{dr}$$

$$\therefore f_R(r) = \begin{cases} 0 & r < d \\ \frac{2r}{l\sqrt{r^2-d^2}} & d \leq r \leq \sqrt{(l/3)^2+d^2} \\ \frac{\cancel{\sqrt{r^2-d^2}}}{\cancel{l}} \cdot \frac{r}{l\sqrt{r^2-d^2}} & \sqrt{(l/3)^2+d^2} < r \leq \sqrt{(2l/3)^2+d^2} \\ 0 & r > \sqrt{(2l/3)^2+d^2} \end{cases}$$

4

For $l = 300$ mile, $d = 20$ mile,

$$F_R(r) = \begin{cases} 0 & r < 20 \\ \frac{2\sqrt{r^2 - 400}}{300} & 20 \leq r \leq \sqrt{10400} = 101.98 \approx 102 \\ \frac{1}{3} + \frac{\sqrt{r^2 - 400}}{300} & \sqrt{10400} < r \leq \sqrt{40400} \approx 201 \\ 1 & r > \sqrt{40400} \end{cases} \quad \text{--- (c)}$$

$$f_R(r) = \begin{cases} 0 & r < 20 \\ \frac{2r}{300\sqrt{r^2 - 400}} & 20 \leq r \leq \sqrt{10400} \\ \frac{r}{300\sqrt{r^2 - 400}} & \sqrt{10400} < r \leq \sqrt{40400} \\ 0 & r > \sqrt{40400} \end{cases} \quad \text{--- (d)}$$

(b) Plot of CDF and PDF,

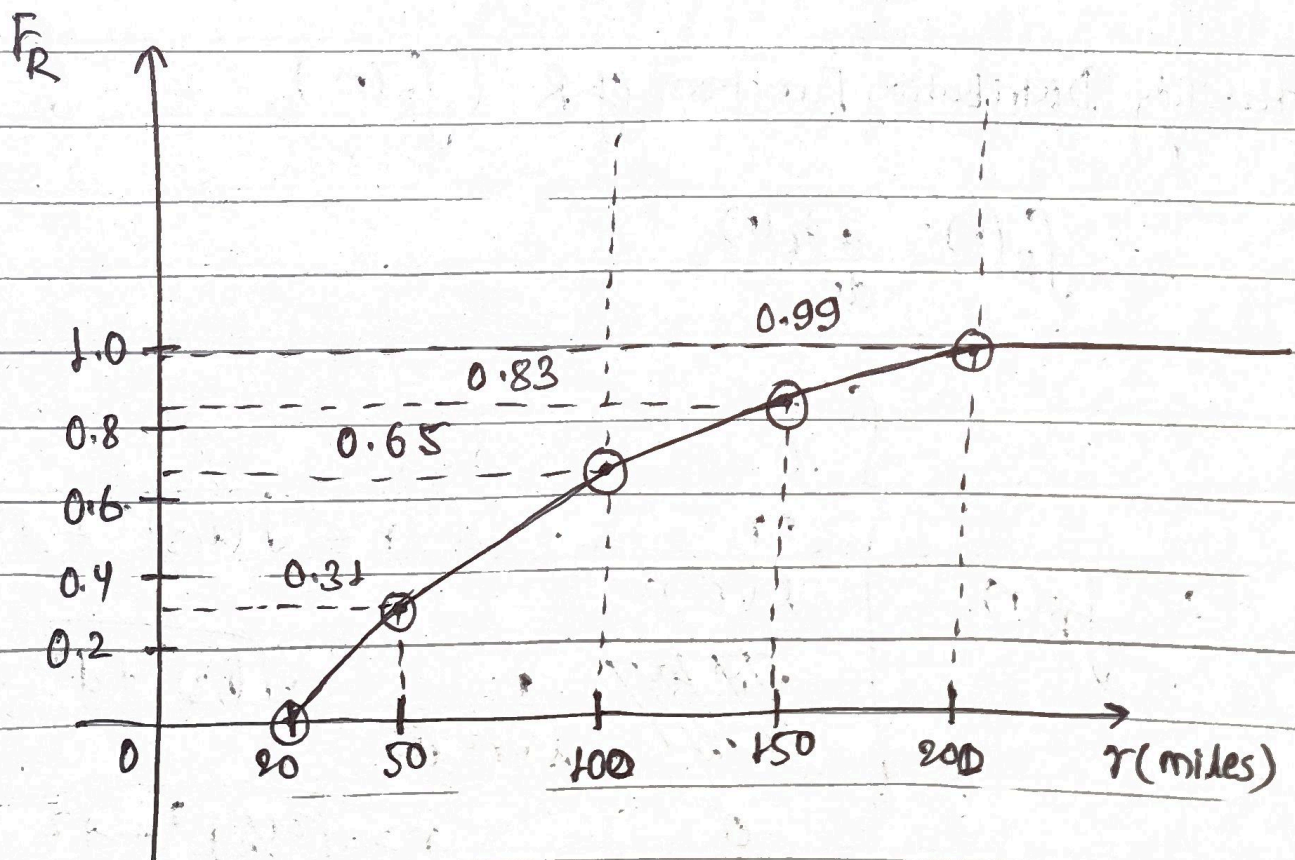


fig. CDF

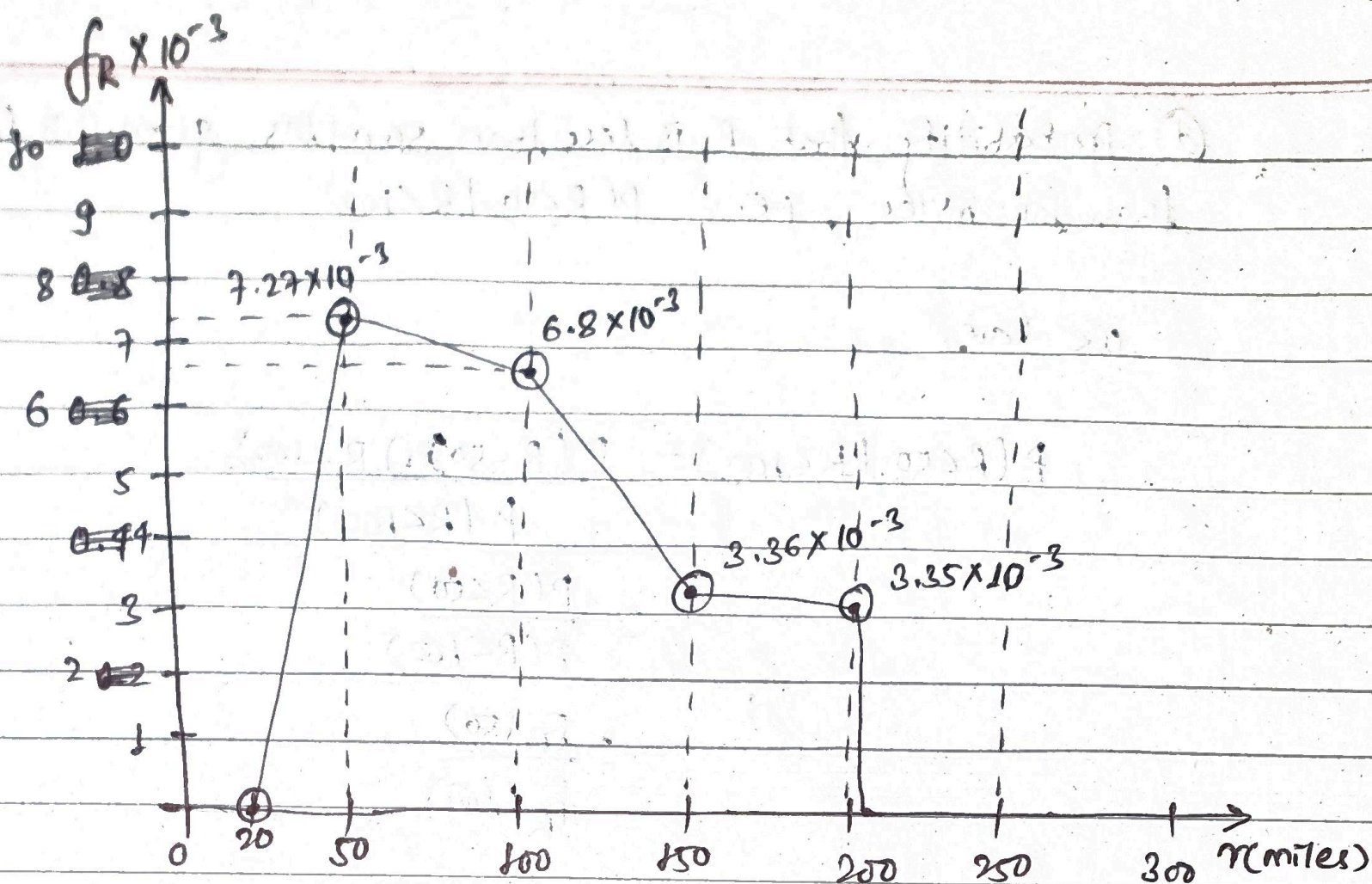


fig. PMF

© Probability that in next earthquake generated by this fault, R is less than or equal to 50 miles,

from eqn ©,

$$P(R \leq 50) \equiv F_R(50) = \frac{2 \sqrt{50^2 - 400}}{300} = 0.3055$$

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④ Probability that R is less than 50 miles given R is less than 100 miles, i.e. $P(R < 50 | R < 100)$

we have,

$$\begin{aligned} P(R < 50 | R < 100) &= \frac{P(R < 50 \cap R < 100)}{P(R < 100)} \\ &= \frac{P(R < 50)}{P(R < 100)} \\ &= \frac{F_R(50)}{F_R(100)} \end{aligned}$$

from eqn ③,

$$P(R < 50 | R < 100) = \frac{F_R(50)}{F_R(100)}$$

$$= \frac{(2/300) \sqrt{50^2 - 400}}{(2/300) \sqrt{100^2 - 400}}$$

$$= 0.4677$$

2

②

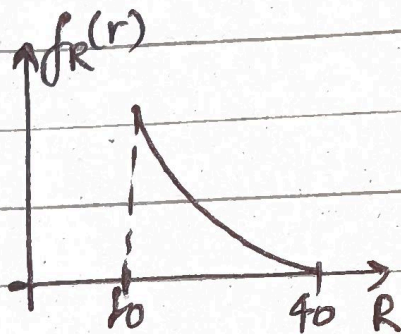
→ Solⁿ:

① $f_R(r) = \frac{a}{r} + b$, and $f_R(r) = 0$ for $r \geq 40$

we have,

At $R=40$, $f_R=0$

$$\Rightarrow 0 = \frac{a}{40} + b \Rightarrow b = -a/40$$



Also,

$$CDF \equiv F_R(r) = \int_{-\infty}^r f_R(r) dr \Rightarrow 1 = \int_{-\infty}^{\infty} f_R(r) dr$$

$$\Rightarrow 1 = \int_{10}^{40} \left(\frac{a}{r} - \frac{a}{40} \right) dr$$

$$\Rightarrow 1 = a \left(\ln r - r/40 \right) \Big|_{10}^{40} = a \left(\ln 40 - 1 - \ln 10 + 0.25 \right)$$

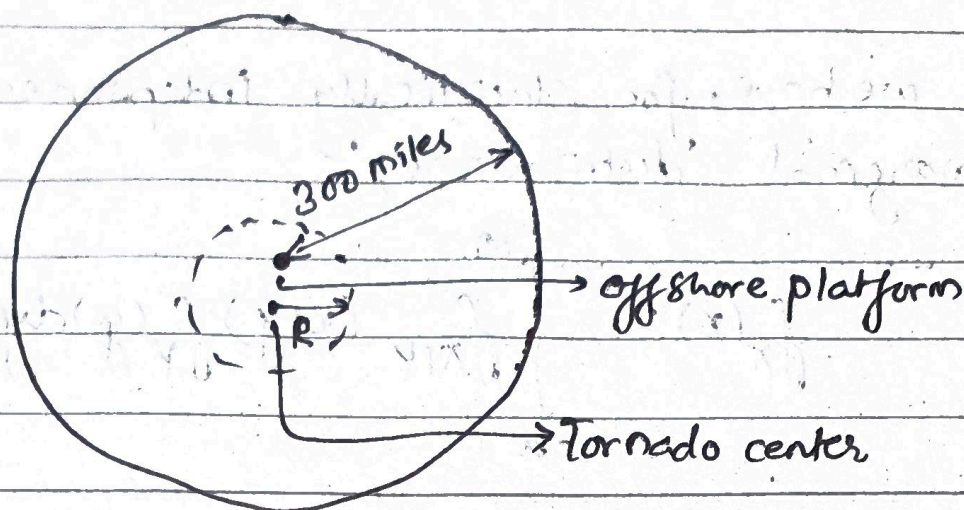
$$= a \left[(3.69 - 1) - (2.30 - 0.25) \right] = a (\ln 4 - 0.75)$$

$$= a (0.64) = a (\ln 4 - 0.75)$$

$$\Rightarrow a = \frac{1}{0.64} = \frac{1}{\ln 4 - 0.75} = 1.5716$$

$$\Rightarrow b = -a/40 = -0.0393$$

⑥ probability that a tornado will damage the offshore platforms



Since the center of tornado is equally likely to occur anywhere within 300 mile of the offshore platform, its PDF is,

$$f_c(c) = \frac{1}{\pi (300)^2} = \frac{1}{90000\pi}$$

Influence area Radius of tornado = R (the offshore platform will be damaged if lying in this area)

Let D be the distance between offshore and tornado center,
The probability that the center lies within distance d when the radius of tornado is r ,

$$\text{i.e. } P(D \leq d | R=r) = \frac{\pi d^2}{\pi (300)^2} = \frac{d^2}{90000}, \quad 0 \leq d \leq 300$$

$$\text{Also, } f_R(r) = \frac{a}{r} = \frac{a}{40}$$

2
for the offshore to be damaged $D \leq r$,

we have, for statistically independent conditional and marginal distributions,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy$$

Thus,

$$\therefore \text{the damage probability} = \int_{-\infty}^{\infty} P(D \leq r | R=r) f_R(r) dr$$

$$= \int_{10}^{40} \frac{r^2}{90000} \left(\frac{a}{r} - \frac{a}{40} \right) dr$$

$$= \frac{a}{90000} \int_{10}^{40} r^2 \left(\frac{1}{r} - \frac{1}{40} \right) dr$$

$$= \frac{a}{90000} \times 225 = \frac{a}{400}$$

$$= \frac{1.5716}{400}$$

$$= 0.003929$$

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 (c) Posterior PDF of R if the offshore has been damaged we have,

$$f''(\theta_i) = \frac{P(E|\theta) f'(\theta)}{\int_{-\infty}^{\infty} P(E|\theta) f'(\theta) d\theta}$$

$$= K L(\theta) f'(\theta)$$

Here,

$$L(r) = \frac{r^2}{90000}, \quad f'_R(r) = \left(\frac{a}{r} - \frac{a}{40} \right) \quad 10 \leq r \leq 40$$

$$L(\theta) = \left[\int_{-\infty}^{\infty} \frac{r^2}{90000} \left(\frac{a}{r} - \frac{a}{40} \right) dr \right]^{-1}$$

$$= \left[\int_{10}^{40} \frac{r^2}{90000} \left(\frac{a}{r} - \frac{a}{40} \right) dr \right]^{-1}$$

$$= \left[\frac{a}{400} \right]^{-1}$$

$$\therefore f''_R(r) = \frac{\frac{r^2}{90000} \left(\frac{a}{r} - \frac{a}{40} \right)}{\frac{a}{400}} = \frac{r^2}{225} \left(\frac{1}{r} - \frac{1}{40} \right)$$

$$\therefore f''_R(r) = \begin{cases} \frac{r^2}{225} \left(\frac{1}{r} - \frac{1}{40} \right) & 10 \leq r \leq 40 \\ 0 & r > 40 \end{cases}$$

③

→ Solⁿ:

Given,

joint PMF of numbers of earthquakes over the two years,

$$P_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{x_1 + x_2}{21} & \text{for } x_1 = 0, 1, 2 \text{ and } x_2 = 2, 3 \\ 0 & \text{elsewhere} \end{cases}$$

The pmf is tabulated as follows,

x_1, x_2	$P_{X_1, X_2}(x_1, x_2)$
0, 2	2/21
0, 3	3/21
1, 2	3/21
1, 3	4/21
2, 2	4/21
2, 3	5/21

① we have marginal PMFs,

$$P_{X_1}(x_{1i}) = \sum_{\text{all } x_{2j}} P_{X_1, X_2}(x_{1i}, x_{2j})$$

$$P_{X_2}(x_{2i}) = \sum_{\text{all } x_{1j}} P_{X_1, X_2}(x_{1j}, x_{2i})$$

$$\therefore P_{X_1}(0) = 2/21 + 3/21 = 5/21$$

$$P_{X_1}(1) = 3/21 + 4/21 = 7/21$$

$$P_{X_1}(2) = 4/21 + 5/21 = 9/21$$

Similarly,

$$P_{X_2}(2) = 2/21 + 3/21 + 4/21 = 9/21$$

$$P_{X_2}(3) = 3/21 + 4/21 + 5/21 = 12/21$$

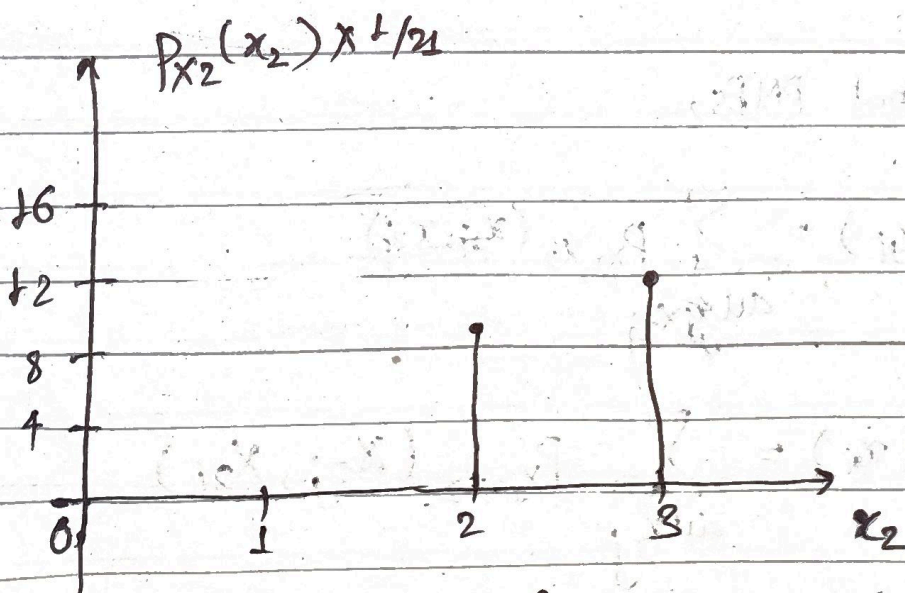
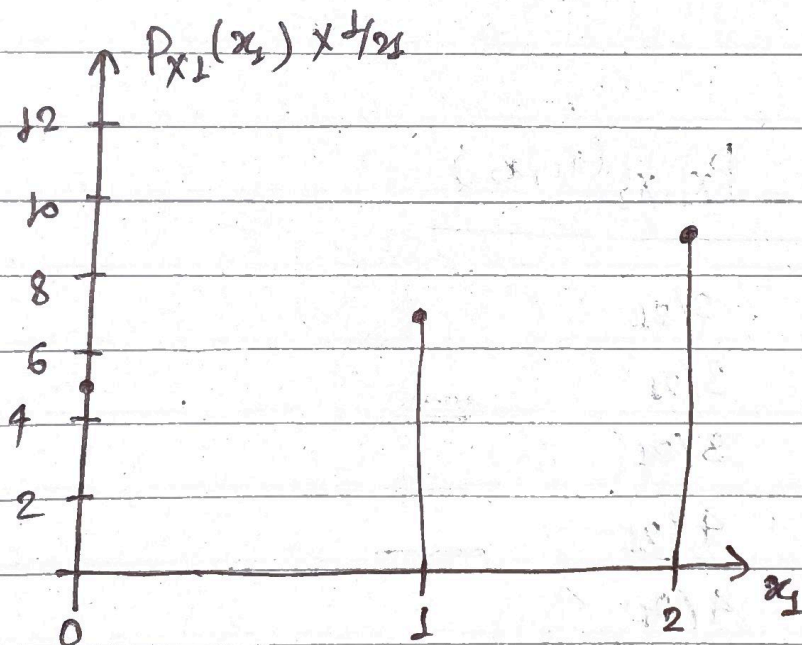


fig. PMF of X_1 and X_2

(b) Mean and Variance of X_1 and X_2 ,

$$E[X] = \sum_{\text{all } x_i} x_i p_X(x_i)$$

$$\Rightarrow E[X_1] = 0 \times \frac{5}{21} + 1 \times \frac{7}{21} + 2 \times \frac{9}{21} = 1.1905 = 25/21$$

$$\text{Also, } E[g(x)] = \sum_{\text{all } x_i} g(x_i) p_X(x_i)$$

$$\Rightarrow E[X_1^2] = \sum_{\text{all } x_i} x_i^2 p_{X_1}(x_i)$$

$$= 0^2 \times \frac{5}{21} + 1^2 \times \frac{7}{21} + 2^2 \times \frac{9}{21}$$

$$= 43/21$$

$$\Rightarrow \text{Var}[X_1] = E[X_1^2] - (E[X_1])^2$$

$$= 43/21 - (25/21)^2 = \frac{278}{441} = 0.6304$$

Similarly,

$$E[X_2] = 2 \times \frac{9}{21} + 3 \times \frac{12}{21} = 18/7 = 2.5714$$

$$E[X_2^2] = 2^2 \times \frac{9}{21} + 3^2 \times \frac{12}{21} = \frac{48}{7} = 6.8571$$

~~$$\text{Var}[X_2] = \frac{48}{7} - \left(\frac{18}{7}\right)^2 = \frac{12}{49} = 0.2449$$~~

$$\text{Var}[X_2] = E[X_2^2] - (E[X_2])^2 = \frac{48}{7} - \left(\frac{18}{7}\right)^2$$

$$= \frac{48}{7} - \frac{324}{49} = \frac{12}{49} = 0.2449$$

14

© Conditional PMF of X_1 given X_2 ,
we have,

$$P_{X_1|Y}(x_i|y_j) \equiv P(X=x_i|Y=y_j) = \frac{P_{X,Y}(x_i, y_j)}{P_Y(y_j)}$$

for $X_2 = 2$,

$$P_{X_1|X_2}(0|2) = \frac{2/21}{9/21} = 2/9$$

$$P_{X_1|X_2}(1|2) = \frac{3/21}{9/21} = 3/9$$

$$P_{X_1|X_2}(2|2) = \frac{4/21}{9/21} = 4/9$$

for $X_2 = 3$,

$$P_{X_1|X_2}(0|3) = \frac{3/21}{12/21} = 3/12$$

$$P_{X_1|X_2}(1|3) = \frac{4/21}{12/21} = 4/12$$

$$P_{X_1|X_2}(2|3) = \frac{5/21}{12/21} = 5/12$$

④ Conditional mean $E[X_1|X_2]$ and Variance $[X_1|X_2]$

$$E[X_1|X_2=2] = 0 \times 2/9 + 1 \times 3/9 + 2 \times 4/9 = 11/9 = 1.22$$

$$E[X_1^2|X_2=2] = 0^2 \times 2/9 + 1^2 \times 3/9 + 2^2 \times 4/9 = 19/9$$

$$\text{Var}[X_1|X_2] = 19/9 - (11/9)^2 = 50/81 = 0.6173$$

K

$$E[X_1 | X_2=3] = 0 \times 3/12 + 1 \times 4/12 + 2 \times 5/12 = 14/12$$

$$E[X_1^2 | X_2=3] = 0^2 \times 3/12 + 1^2 \times 4/12 + 2^2 \times 5/12 = 24/12 = 2.667$$

$$\begin{aligned} \text{Var}[X_1 | X_2=3] &= 24/12 - (14/12)^2 = 23/36 \\ &= 0.639 \end{aligned}$$

g) Unconditional mean $E[X_1]$ and variance $[X_1]$ using results from d),

$$\text{we have, } E[X_1] = \sum_{\text{all } x_2} E[X_1 | X_2=x_2] \cdot p_{X_2}(x_2)$$

$$\begin{aligned} E[X_1] &= \frac{11}{9} \times \frac{9}{21} + \frac{14}{12} \times \frac{12}{21} \\ &= \frac{25}{21} \end{aligned}$$

Similarly, expected value of variance is,

$$\begin{aligned} \text{Var}[X_1] &= \frac{50}{81} \times \frac{9}{21} + \frac{23}{36} \times \frac{12}{21} \\ &= \frac{17}{27} \end{aligned}$$

$$= 0.6296$$

~~which~~ Thus, $E[X_1]$ is same as in question (b),
from (b) $\text{Var}[X_1] = 0.6304$ which is $\approx \text{Var}[X_1]$ obtained here.

④

→ Sol?

we have,

for a given function, $Y = g(X)$ first order ~~the~~ approximate mean and variance of Y as,

$$E(Y) \approx g(\mu_x)$$

$$\text{Var}(Y) \approx \text{Var}(X) \left(\frac{dg}{dx} \right)_{\mu_x}^2$$

Here,

$$D = \frac{5\omega L^4}{384EI}$$

Where,

$$L = 15 \text{ ft} = 15 \times 12 = 180 \text{ in}$$

$$\omega = 1000 \text{ lb/ft} = 1000/12 = 83.33 \text{ lb/in}$$

$$E \Rightarrow \text{mean} = 3,000,000 \text{ psi}, \text{cov} = 25\%$$

$$\text{Dimension of cross-section} = 6 \text{ in} \times 12 \text{ in}$$

$$\therefore I = \frac{bh^3}{12} = \frac{6 \times 12^3}{12} = 864 \text{ in}^4$$

$$\begin{aligned} \therefore E(D) &\approx \frac{5 \times 83.33 \times 180^4}{384 \times (3,000,000) \times 864} \\ &= 0.4394 \text{ in} \end{aligned}$$

$$\begin{aligned} \text{Var}(D) &= \sigma_E^2 = (0.25 \times \mu_E)^2 = (0.25 \times 3,000,000)^2 \\ &= (750,000)^2 \end{aligned}$$

$$\text{Var}(D) \approx \text{Var}(E) \left(\frac{dD}{dE} \right)^2$$

$$= (750\,000)^2 \left(\frac{5014}{3841} \times \frac{(-1)}{\mu_x^2} \right)^2$$

$$= (750\,000)^2 \left(\frac{.5 \times 83.33 \times 180^4}{384 \times 864} \times \frac{-1}{3000\,000^2} \right)^2$$

$$= 0.012069$$

$$\Rightarrow \sigma_D = \sqrt{0.012069} \\ = 0.1099 \text{ m}$$

⑥ The second-order mean is given by,

$$E(Y) \approx g(\mu_x) + \frac{1}{2} \text{Var}(x) \frac{d^2 g}{dx^2}$$

Here,

$$E(D) \approx 0.4394 + \frac{1}{2} (750\,000)^2 \frac{d}{dE} \left(\frac{5014}{3841} \frac{-1}{E^2} \right)$$

$$= 0.4394 + \frac{1}{2} (750\,000)^2 \left(\frac{5014}{3841} \right) \frac{(-1)(-2)}{E^3}$$

$$= 0.46686$$

$$= 0.4669$$

(b) The second order mean

we have,

$$\text{for } Y = g(X)$$

Second order approximation is,

$$E(Y) \approx g(\mu_X) + \frac{1}{2} \text{Var}(X) \left. \frac{d^2 g}{dX^2} \right|_{\mu_X}$$

Here,

$$E(D) \approx \frac{5 \times \left(\frac{1000}{12} \right) \times 180^4}{384}$$

$$E(D) \approx \frac{5 \omega l^4}{384 \mu_E I} + \frac{1}{2} \text{Var}(E) \left. \frac{d^2 D}{dE^2} \right|_{\mu_E}$$

$$\frac{d^2 D}{dE^2} = \frac{d^2}{dE^2} \left(\frac{5 \omega l^4}{384 E I} \right) = \frac{5 \omega l^4}{384 I} \frac{d^2}{dE^2} \left(\frac{1}{E} \right) \Big|_{\mu_E}$$

$$= \frac{5 \omega l^4}{384 I} \frac{2}{E^3} \Big|_{\mu_E} = \frac{5 \omega l^4}{384 I} \left(\frac{2}{\mu_E^3} \right)$$

$$\therefore E(D) \approx \frac{5 \omega l^4}{384 \mu_E I} + \frac{1}{2} \text{Var}(E) \frac{5 \omega l^4}{384 I} \frac{2}{\mu_E^3}$$

$$= 0.4394 + \frac{1}{2} (750000)^2 \frac{5 \times \frac{1000}{12} \times 180^4 \times (2)}{384 \times 864 \times (3000000)^3}$$

$$= 0.46692 \text{ m}$$

© If E is lognormal with parameters λ_E, ξ_E^2 then

$\ln E$ is normally distributed as $N(\lambda_E, \xi_E^2)$

Here,

$$D = \frac{5\omega l^4}{384EI} = \frac{C}{E} \quad \text{where, } C = \frac{5\omega l^4}{384I}$$

$$C = \frac{5 \times \frac{1000}{12} \times 180^4}{384 \times 864}$$

$$= 1318359.375$$

$$D = C/E$$

taking log on both sides,

$$\ln D = \ln C/E = \ln C - \ln E = (-1) \ln E + \ln C$$

$$\equiv Y = aX + b$$

Since $\ln E$ is normally distributed, $\ln D$ is also normally distributed. Thus D is lognormal.

Lognormal Parameters for E :

$$\xi_E^2 = \ln(1 + \delta_E^2)$$

$$= \ln(1 + 0.25^2) = 0.060625$$

$$\lambda_E = \ln \mu_E - \frac{1}{2} \xi_E^2 = \ln(3000000) - \frac{1}{2} \times 0.060625$$

$$= 14.883810$$

$$\therefore E \Rightarrow \text{lognormal}(\lambda_E, \xi_E^2) = \text{lognormal}(14.88, 0.0606)$$

$$\ln E \Rightarrow N(\lambda_E, \xi_E^2) = N(14.88, 0.06062)$$

lognormal parameters for D,

for a linear function $Y = aX + b$,

$$E(Y) = a E(X) + b$$

$$\text{Var}(Y) = a^2 \text{Var}(X)$$

$$\text{for } \ln D = (-1) \ln E + \ln C$$

$$\xi_D^2 = \text{Var}(\ln D) = (-1)^2 \text{Var}(\ln E) = \xi_E^2 = 0.060625$$

$$\lambda_D = \cancel{E(D)} E(\ln D) = (-1) E(\ln E) + \ln C$$

$$= (-1) 14.88381 + \ln(1318359.375)$$

$$= -0.791912$$

$$\therefore \ln D \Rightarrow N(-0.7919, 0.0606)$$

$$D \Rightarrow \text{lognormal}(-0.7919, 0.0606)$$

we have, for lognormal Distribution,

$$f_X = \frac{1}{\sqrt{2\pi} (\xi_X)} \exp. \left[-\frac{1}{2} \left(\frac{\ln X - \lambda}{\xi} \right)^2 \right] \quad x \geq 0$$

$$\therefore f_D(d) = \frac{1}{\sqrt{2\pi} \xi_D d} \exp \left[-\frac{1}{2} \left(\frac{\ln d - \lambda_D}{\xi_D} \right)^2 \right], \quad d \geq 0$$

$$\sqrt{2\pi} \xi_D = \sqrt{2\pi} \times \sqrt{0.060625} = 0.6172$$

$$\Rightarrow f_D(d) = \frac{1}{0.6172 d} \exp \left[-\frac{1}{2} \left(\frac{\ln d + 0.7919}{0.24622} \right)^2 \right], \quad d \geq 0$$

$$= \frac{1}{0.6172 d} \exp \left[-\frac{(\ln d + 0.7919)^2}{0.12125} \right], \quad d \geq 0$$

Exact Mean And Coefficient of variation:

we have,

$$\xi_D^2 = \ln(1 + \delta_D^2) \Rightarrow \delta_D = (e^{\xi_D^2} - 1)^{1/2}$$

$$\therefore \delta_D = (e^{0.060625} - 1)^{1/2} = 25\%$$

$\therefore$ exact coefficient of variation of D = 25%.

Also,

$$\lambda_D = \ln \mu_D - \frac{1}{2} \xi_D^2 \Rightarrow \mu_D = \exp \left(\lambda_D + \frac{1}{2} \xi_D^2 \right)$$

$$\Rightarrow \mu_D = \exp \left(-0.791912 + \frac{1}{2} \times 0.060625 \right)$$

$$= 0.46692 \text{ m}$$

$\therefore$ exact mean = 0.46692 m